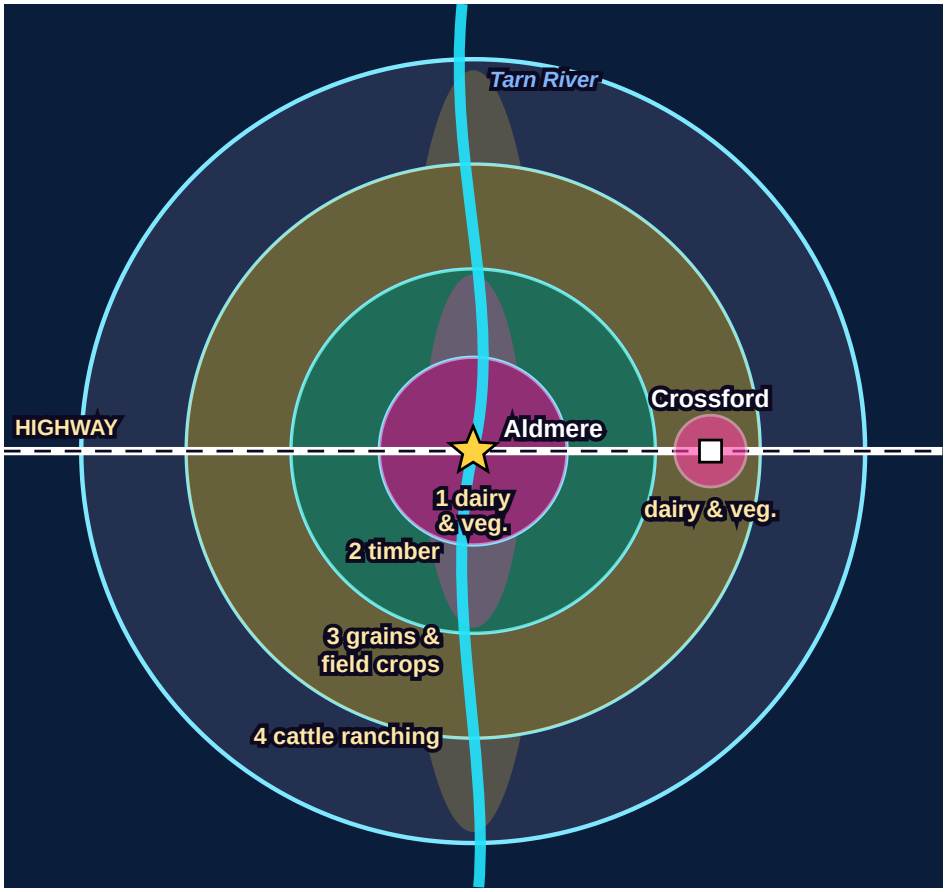
Rings of Farmland

AP Human Geography · Unit 5 · One stimulus (like Question 2) · 7 points · Teacher key · GeoArcade FRQ Tutor

This diagram shows land use around a market city in a fictional farming region. Farmers sell their products in the city of Aldmere. A river flows through the region, and a highway crosses it from west to east.

Agricultural Land Use Around Aldmere, a Fictional Market City



1 Dairy & vegetables · 2 Timber / forest · 3 Grains & field crops · 4 Cattle ranching · River · Highway / town · Market city

Diagram created for practice. Aldmere, Crossford, and the Tarn River are fictional. Simplified and not to scale; lighter shading shows where a ring is stretched.

A. Identify the agricultural activity found in ring 1, the ring closest to the market city. IDENTIFY

EARNS THE POINT IF THE ANSWER DOES ANY ONE OF THESE:

- Names dairy farming or vegetable/market gardening
- Describes it as perishable or intensive farming near the city

MODEL ANSWER

Ring 1 is dairy farming and market gardening (growing fresh vegetables) for the city.

Common mistake: “Farms.” Every ring is farms. Identify means name the specific activity: dairy and vegetable (market) gardening.

Why the model works: It names the exact activity on the label (dairy and vegetables) and uses the geographer's term for it (market gardening).

B. Explain why the agricultural products grown closest to the market are the most perishable or the most expensive to transport. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names the trait: perishable (spoils fast) or heavy/bulky products like milk, vegetables, or timber
- Explains the logic: transport cost rises with distance, so these farmers gain the most from being close and can outbid others for near-city land (bid-rent)

MODEL ANSWER

T1- Products that spoil quickly or are heavy to haul are grown closest to the market because transport cost rises with distance from the city.
E2- Milk and fresh vegetables can rot within a day or two of harvest, and timber is bulky and heavy to carry by wagon, while grain keeps for months and cattle can walk to market.
A3- Because dairy farmers and market gardeners lose the most money from a long trip, they gain the most from being near the city and can afford to pay (bid) more for the expensive land there, which pushes grain and ranching farther out.

Common mistake: “Farmers near the city grow what city people want to buy.” That doesn't explain distance. City people buy bread and beef too, but grain and cattle come from far away. You need the transport-cost and spoilage logic.

Why the model works: It states the rule (transport cost rises with distance), gives specific products as proof (milk, vegetables, timber vs. grain and cattle), and explains the bidding result (perishable-product farmers outbid others for near-city land).

C. Using the diagram, describe how the river changes the pattern of land use compared with the classic von Thünen model. DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Describes the change with diagram evidence: rings (1 and 3) are stretched/elongated into ovals along the river instead of perfect circles
- Links it to cheaper or faster transport along the river (boats/barges lower transport cost)

MODEL ANSWER

T1- In the classic model the rings are perfect circles, but on this diagram the river stretches the rings into long ovals.
E2- The lighter shading shows ring 1 (dairy and vegetables) extending north and south of Aldmere along the Tarn River into the timber ring, and ring 3 (grains) extending along the river into the ranching ring.
A3- This happens because moving goods by boat along the river is cheaper and faster than hauling them by wagon, so a farm that is far away in miles but on the river is “close” in transport cost and can grow the same products as farms next to the city.

Common mistake: “The river gives the farms water, so farmers near the river can grow more crops.” That's irrigation, not the model. The question is about the SHAPE of the rings. Describe how they are stretched along the river and connect it to cheaper transport by boat.

Why the model works: It contrasts the classic circles with the stretched ovals, cites the exact rings and direction from the diagram, and explains the cause (cheaper boat transport makes river farms “closer” in cost).

D. Explain how the secondary market town of Crossford along the highway shows how modern transportation changes the von Thünen model. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Identifies what Crossford shows: a second market with its own mini dairy/vegetable zone (ring), breaking the one-market assumption
- Explains the role of modern transport: highways/trucks make travel faster and cheaper, so markets and rings form in new places

MODEL ANSWER

T1- Crossford shows that modern transportation breaks the model's assumption of a single market with one set of rings.
E2- On the diagram, a small dairy-and-vegetable zone surrounds Crossford where the highway crosses ring 3, forming a miniature ring 1 inside the grain zone.
A3- Because trucks on the highway move goods quickly and cheaply, Crossford grew into a second market of its own, so the region now has overlapping sets of rings instead of one neat set of circles around Aldmere.

Common mistake: “The highway makes it easier for people to drive to Aldmere.” That's about commuters, not land use. The question wants what Crossford shows about the model: a second market with its own mini ring, created because trucks on the highway cut transport cost.

Why the model works: It names the broken assumption (one market), uses diagram evidence (the mini dairy zone around Crossford on the highway), and explains the cause (fast, cheap trucking creates a second market and overlapping rings).

E. Explain ONE way refrigerated trucking and global shipping have weakened the von Thünen pattern at the national or global scale. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names the technology: refrigeration/cold chain, refrigerated trucks, container ships, air freight, highways
- Explains the effect with scale: perishables are grown far away (e.g., Chile, Mexico, California) so the near-city rings weaken at the national or global scale

MODEL ANSWER

T1- Refrigeration has weakened the model because perishable products no longer have to be grown next to the city that eats them.

E2- Refrigerated trucks and container ships keep produce fresh for weeks, so a U.S. supermarket in January sells grapes from Chile and tomatoes from Mexico that traveled thousands of miles.

A3- At the national and global scale, this means the ring 1 logic breaks down: the dairy-and-vegetable zone is not a ring around each city but far-off regions like California, Florida, or another country, so the whole world acts as one giant hinterland.

Common mistake: “Trucks are faster than horses, so people can sell more.” That's true but doesn't explain the model. Name the technology (refrigeration), give an example (Chilean grapes in January), and explain how it breaks the near-city ring at the national or global scale.

Why the model works: It names the technology (refrigerated trucks and container ships), gives a concrete example with distance (Chile, Mexico, thousands of miles), and explains the effect at a named scale (national/global rings dissolve).

F. Describe ONE environmental consequence of intensive commercial agriculture in ring 1 or ring 3 of the diagram. DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a specific environmental consequence: fertilizer/pesticide/manure runoff, water pollution, aquifer or groundwater depletion, soil salinization, soil erosion, or loss of soil fertility
- Describes how it happens or what it does to the environment (cause → result)

MODEL ANSWER

T1- One environmental consequence of intensive farming in ring 3 is water pollution from fertilizer runoff.

E2- Grain farmers spread large amounts of nitrogen fertilizer on their fields, and when it rains the extra fertilizer washes off the fields and flows into the Tarn River.

A3- The nitrogen feeds huge algae blooms downstream that use up the oxygen in the water and kill fish, creating a "dead zone" like the one at the mouth of the Mississippi River in the Gulf of Mexico.

Common mistake: "Farming in ring 3 uses up a lot of land." Using land isn't an environmental consequence by itself. Name a specific harm (fertilizer runoff, aquifer depletion, salinization) and describe how it happens.

Why the model works: It names a specific consequence (fertilizer runoff), describes the mechanism (rain washes nitrogen into the river), and gives the result with a real example (algae, dead zones, Gulf of Mexico).

G. Explain ONE limitation of using this model to understand agriculture in a less developed country today. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names an assumption that doesn't fit: subsistence (not commercial) farming, no single market / many small markets, export crops for ports, rugged terrain or varied climate, land tenure (communal land, plantations, landless farmers)
- Explains why that makes the model misleading there

MODEL ANSWER

T1- One limitation is that the model assumes commercial farmers who sell everything to a single market city, which does not describe many farmers in less developed countries.
E2- In rural parts of countries like Ethiopia or Bolivia, many families practice subsistence agriculture, growing maize, teff, or potatoes mainly to feed themselves, often on steep hillsides or communal land rather than a flat plain with one market.
A3- Because those farmers are not choosing crops based on transport cost to a central city, distance to the market does not sort their land use into rings, so the model would predict the wrong pattern and miss what actually shapes their farming, such as rainfall, terrain, and land ownership.

Common mistake: “The model is old, so it doesn't work anymore.” Being old isn't a limitation by itself; the transport-cost logic still works in many places. Name a specific assumption that fails in a less developed country (subsistence farming, no single market, terrain, land tenure) and explain why that matters.

Why the model works: It names the failing assumption (commercial farmers and a single market), gives specific detail (subsistence farming on hillsides and communal land in Ethiopia or Bolivia), and explains the consequence (distance doesn't sort land use, so the model predicts wrongly).
